BAN404: Statistical Learning

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What is business analytics?

- Let us first ask what *analytics* means. According to Davenport and Harris (2007), it means
- "the extensive use of data, statistical and quantitative analysis, explanatory and predictive models and fact based management to drive decisions and actions"
- *Business analytics* is when we put into the context of how firms are using their available data to make the best possible decisions in the most efficient way.

Three categories of business analytics

- *Descriptive analytics*: Descriptive/explorative data analysis to understand the past and the present. An example of this is to investigate the salary structure in a company. This is not based on a particular model, nor is it prescribing any actions. It is simply a statement of facts.
- $\boxed{\textit{Predictive analytics}}$: Forecasts of future events, e.g. tomorrows price of the IBM-stock, or prediction of an unknown property, e.g. estimating the probability that an individual has evaded taxes.
- *Prescriptive analytics*: Based on an optimization model or a statistical analysis we say what should be done, what is the best decision. Examples of this are to allocate vehicles in a transportation network or a clinical trial where the purpose is to decide which medication is the best to use for patients with a particular disease.

Three essential components to solve a business analytics problem

- Management Science / Operations Research: Formulating and solving a decision problem for a well specified real world business question (e.g., Which potential customers should be targeted in a marketing campaign?) and transforming it into an actionable form.
- $\boxed{\text{Data analysis / Statistics}}$: Which data and which data analytic methods should be used to estimate or predict quantities of interest (e.g., the probability of a potential customer responding postively to the marketing campaign).
- Programming / Database management: Competence in order to implement, at least a prototype of, a solution.

Recommended preparation to lectures

An Introduction to Statistical Learning by Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani

1. Read the relevant chapter in the book
2. Make sure that you have, and are able to read, the data that will be analysed on the lecture, into R
3. If time permits, try to start solving the exercise(s) for the lecture

Tutorials and groups

- Form groups, of preferably 4 students, for the tutorials and the compulsory assignment.
- During the 4 tutorials you will work together on exercises connected to the previous lecture

Examination

- Digital school exam (6 hours) where R will be used.
- Focus will be on understanding and implementation of methods and on applications to real data.

R and Rstudio

- R
- Rstudio

Statistical Learning

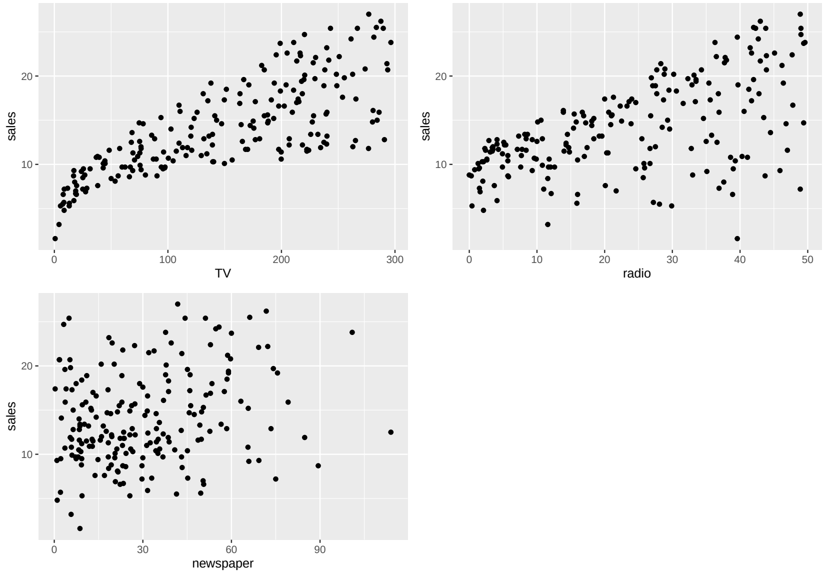
Example

Improve sales on a product

Inputs, explanatory, independent variables:
$X_1 =$ TV budget
$X_2 =$ Radio budget
$X_3 =$ Newspaper budget

Output, response, dependent variable
$Y =$ Sales

Advertising example



Estimation of f

- If one activity should define this course, this is it. The scope of this is, however much broader than it initially looks like.
- The purpose is either
  - Inference
  - Prediction (which we will primarily focus on)

Inference

Answer the questions

1. Which predictors are associated with the response? Or even, which predictors *cause* the response.
2. How strong is the association for different predictors? (Variable importance)
3. What is the functional form, i.e., what is the shape of the function f?

Inference in advertisement example

1. Which media contribute to sales? Can the association between the X's and Y be used to increase sales by changing budgets
2. Which media have most impact on sales?
3. Is the effect of an increase of, say, TV ads on sales linear

Supervised and unsupervised learning

- Supervised learning is when we observe Y
- **Example**: We have a training data where we know individuals that have evaded taxes.
- Unsupervised learning is when we do not observe Y
- **Example**: We have to look for "remarkable" observations, e.g. individuals who we observe consume much more than their declared income and wealth would suggest.

Setup for most methods treated in this course

- Generally: $\mathbf{X} = (X_1, ..., X_p)'$
- $Y = f(\mathbf{X}) + \varepsilon$, is the Data Generating Process (DGP).
- As the name indicates, this is the way data has been generated. This is, obviously not known before the analysis starts and one can only hope to find a reasonable approximation to it after a lot of work.
- f is a fixed function which is unknown. Represents systematic information of Y contained in $\mathbf{X}$.
- ε is a zero mean random error term, independent of $\mathbf{X}$

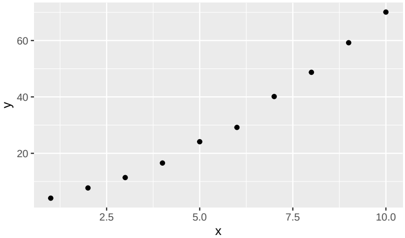
Prediction

- $Y = f(\mathbf{X}) + \varepsilon$
- $\hat{Y} = \hat{f}(\mathbf{X})$ where $\hat{f}$ is an estimator of f.
- Simplest example: DGP $Y = \alpha + \beta X + \varepsilon$,
  i.e. $\hat{Y} = \hat{f}(X) = \hat{\alpha}_{OLS} + \hat{\beta}_{OLS} X$
- *Reducible error*: $\hat{f}$ not a perfect estimator of f; goes away with large n.
- *Irreducible error*: Y is also dependent on ε; does not.

Overfitting

- Think about linear regression - if an additional X-variable is added, R^2 will, without exception, increase; even though the variable is completely useless as a predictor for Y.
- Consider the the following *training* data:

x	y
1	4.09
2	7.71
3	11.39
4	16.55
5	24.11
6	29.18
7	40.13
8	48.72
9	59.22
10	70.08



- We call this *training* data since we use it to train, to estimate, our model.
- Predict Y for $X = 15$

Linear regression

x	y
1	4.09
2	7.71
3	11.39
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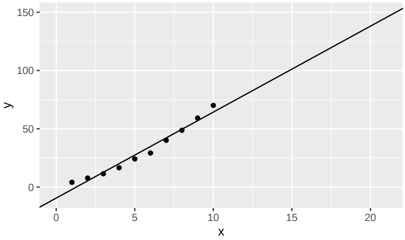
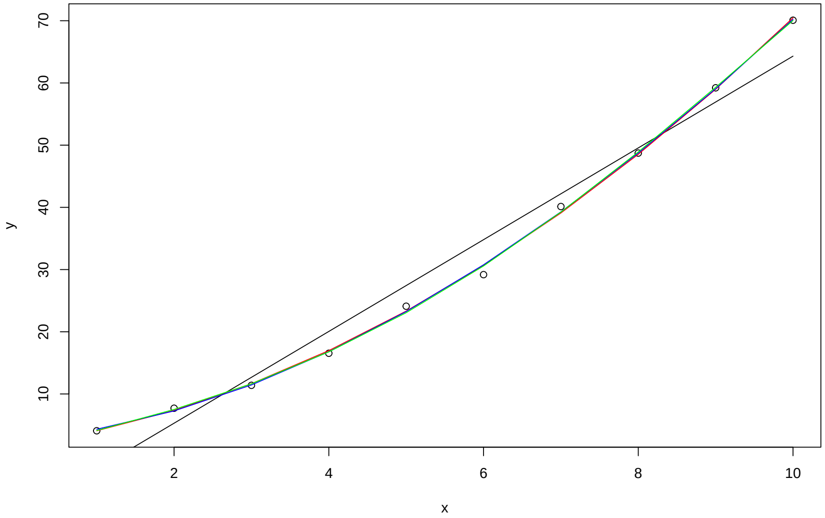
9	59.22
10	70.08



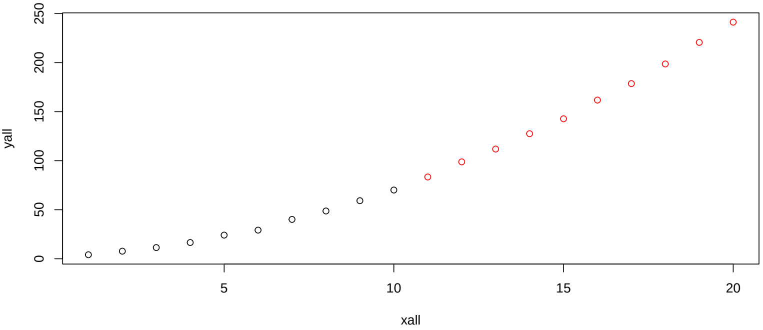
- Estimated model: $\hat{Y} = -9.45 + 7.38X$
- Prediction for $X = 15$: $\hat{Y} = 101.19$

Polynomial regression

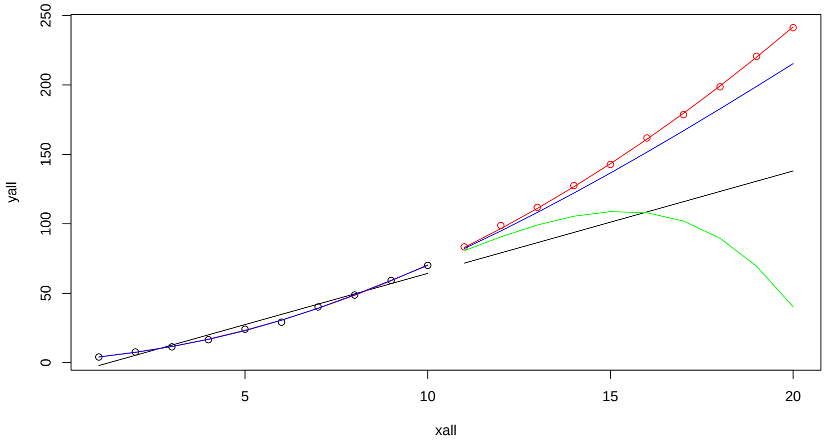
$$Y = \alpha_0 + \alpha_1 X + \alpha_2 X^2 + ... + \alpha_k X^k + \varepsilon \text{ for } k = 1, 2, 3, 4.$$



What if we consider other data

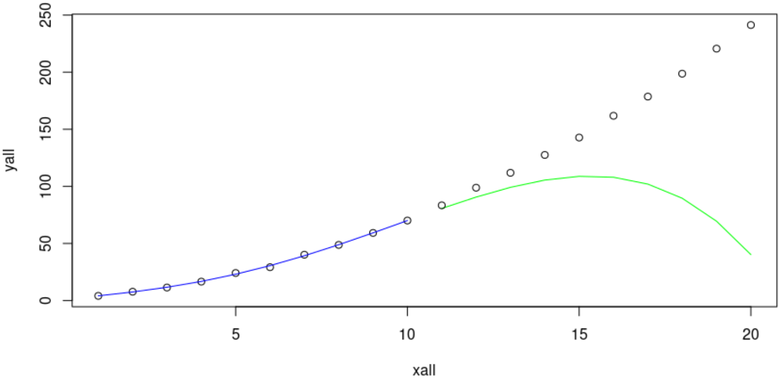
- We actually have 20 observations and we call the last 10 our *test* data. We use this term since we are using this data to test our models.
- 

What if we consider other data



- The linear model is *underfitting* the data and the 3rd and 4th order polynomials are *overfitting* it. How can we deduce that the data was generated by a 2nd order polynomial?

Overfitting



4th order poynomial fitted to observations 1-10, training data

fitted model used to predict observations 11-20, test data

Fitting functions - predict with a model that is a constant (a very, very simple model, linear regression of Y on a constant)

Assume Y is a stochastic variable with cumulative distribution function (cdf) F and probability density function (pdf) f, expectation μ and standard deviation σ. Find θ that minimizes

- $$g(\theta) = E((Y-\theta)^2).$$

$$g(\theta) = \int_{-\infty}^{\infty} (y-\theta)^2 f(y)dy = \int_{-\infty}^{\infty} (y^2 - 2\theta y + \theta^2) f(y)dy$$

$$= \underbrace{\int_{-\infty}^{\infty} y^2 f(y)dy}_{E(Y^2)=Var(Y)+E(Y)^2} - 2\theta \underbrace{\int_{-\infty}^{\infty} y f(y)dy}_{E(Y)} + \theta^2 \underbrace{\int_{-\infty}^{\infty} f(y)dy}_{1}$$

$$= \sigma^2 + \mu^2 - 2\theta\mu + \theta^2$$

Fitting functions

$$g(\theta) = \sigma^2 + \mu^2 - 2\theta\mu + \theta^2$$
$$g'(\theta) = -2\mu + 2\theta$$
$$g'(\theta) = 0 \Longrightarrow \theta_{opt} = \mu = E(X)$$
$$g''(\theta_{\min}) = 2 > 0 \Longrightarrow \theta_{\min} = \mu$$

Fitting functions

- $\theta = E(Y)$ is minimizing $E((Y-\theta)^2)$
- The corresponding empirical quantity to $E((Y-\theta)^2)$ is $\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\theta})^2$
- which is minimized by $\hat{\theta} = \overline{y}$

Another fitting function

- $$g(\theta) = E(|Y - \theta|).$$
- Minimizing this fitting function leads to another "prediction"
$$\theta_{\text{min}} = \text{median}(Y)$$
- Other loss (fitting) functions, e.g., financial loss, can lead to other optimal predictions
- The same model can, thus, give different optimal predictions for different loss functions

Another model

- In the previous example we started with the unconditional distribution of the stochastic variable Y and tried to predict with the "best" constant given a fitting function.
- Consider now instead that we belive that X can help in predicting Y according to the linear model

$$Y = \alpha + \beta X + \varepsilon,$$

  where $\varepsilon \sim N(0, \sigma^2)$
- This implies that the conditional distribution, $(Y|X = x) \sim N(\alpha + \beta x, \sigma^2)$
- With an argument, very similar to the one above, we can deduct that the value of θ that minimizes $E((Y - \theta)^2|X = x)$ is $\theta_{min} = E(Y|X = x) = \alpha + \beta x$.
- What if we minimized $E(|Y - \theta|\,|X = x)$ instead?

A more general model

- What if the model is $Y = f(X) + \varepsilon$?
- For a given $X = x$, $E(\varepsilon^2)$ is minimized by $E(Y|X = x) = f(x)$
- $E((Y - f(X))^2)$ and $E(|Y - f(X)|)$, two possible fitting functions. Any other possibilities. Financial rewards, classification problems...